

LIGHTWEIGHT, HIGH-BANDWIDTH CONFORMAL ANTENNA SYSTEM FOR BALLISTIC HELMETS

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ABSTRACT

Current military radio antennas for man-portable radios suffer from bandwidth, pattern, overall gain limitation and covertness. A number of helmet antenna designs will be analyzed and demonstrated in the bands of interest of 750-870 MHz, and 1350-2700 MHz. The helmet antenna solutions use conductive ink printing techniques to realize antenna elements that are conformal to the helmet's shape. The solutions are highly integrated into the helmet and other equipment on the soldier head for a helmet antenna system with negligible or no equipment weight gain. The design of the antenna system will take into account of the electromagnetic coupling of the head, torso and other soldier equipment on the performance of the antenna system. A Specific Absorption Rate (SAR) study will ensure safety for the user.

INTRODUCTION

The Future Force Warrior (FFW) Advanced Technology Demonstration (ATD) made the assessment that helmet antenna technology represents high risk for the FFW and decided on body worn antennas without any helmet antenna components. However, given conformal helmet antenna's low visual signature and potential for improved performance, the US Army has funded SI2 Technologies and their partner L-3 Communications-CSW to investigate solutions that mitigate the technical concerns and enable helmet antennas to be a practical solution for the future soldier combat system beyond the FFW ATD.

The Advanced Combat Helmet (ACH) is an ideal location for Radio Frequency (RF) communication antennas given that it is the highest point on the soldier with minimal co-located interferers. A body worn antenna that is necessarily at a lower height, has the potential of interference with other gear hanging off the soldier. Performance of the body worn antenna is also highly dependent on the soldier's fighting position, such

as below ground level in a fox hole or on the ground in a prone position. Additionally, there are many soldier system configuration variations employing weapons, ammunition, communication equipment, and sustainment supplies all mounted below the neck to complicate any fielding of body worn antennas. The helmet antenna also has the advantage of being on a rigid, stable platform located high on the soldier combat system away from many interferers.



Figure 1. Broad-band antenna and ground plane written on ACH helmet and under camouflage.

Presently, the major areas of concern in the design of helmet antennas are size and weight, a cumbersome and inflexible antenna connection to the radio (the radio is normally carried on the torso of the soldier), Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC) with other electronic equipment (such as headsets, displays, and weapon targeting and acquisition systems), and health issues associated with RF radiation absorbed by human tissue.

In this paper we will present an antenna architecture and system design of an integrated helmet antenna solution, for the bands of 750-870 MHz and 1350-2700 MHz, that will address these areas of concern. The manufacturing challenges and philosophy for integrating the antenna within the helmet structure will be discussed, followed by a description of a helmet antenna prototype and test results.

MULTIPLE HELMET ANTENNA APPLICATIONS

The applications that envisioned for the helmet antenna are voice communication, direction finding (DF), jamming, situation awareness (SA), GPS, remote sensor data and video. These applications would require a range of bandwidth. Although the surface area on the helmet is rather limited, a broad band helmet antenna system may still be able to replace multiple, single use antennas. It is envisioned that, with the addition of integrated conformal helmet antennas, an system architecture with on-demand configurable apertures working with an advanced application management system will enable the soldier to have a single, user-transparent antenna system that would satisfy all RF needs.

ANTENNA ARCHITECTURE AND SYSTEM DESIGN

The range of the bands of interest is 750-2700 MHz or 1950 MHz of bandwidth. The center frequency is 1725 MHz requiring a bandwidth of $1950/1725 = 113\%$. The impedance bandwidth of an antenna is commonly determined by the range where the VSWR is less than a specific value, in this case 2.5:1. Very few antenna architectures can achieve such high bandwidths. Patch antennas for instance, often have a bandwidth of only a few percent. The genetically optimized monopole described in [1] has the potential to meet the bandwidth requirement. In [2], the idea was extended to a single printed circuit board (PCB) with a planar architecture. A ground plane is placed on the substrate, with the antenna

element on a parallel layer. The antenna is fed with a microstrip line that is fed by a connector at the bottom of the antenna.

The planar architecture is readily transferable to SI2's direct write technology using conductive inks. However, in this case, the antenna and ground plane are on the same layer. A thin dielectric is printed on the ground plane and a microstrip line is printed on the dielectric. Direct write can be used to print on planar Kevlar surfaces and conformal surfaces such as ballistic helmets as shown in Figure 1. The antenna's length is determined by the lowest frequency, and good impedance matching is possible with a total length of less than 0.4 wavelengths.

Minimizing the weight is a key performance metric of this effort. SI2's direct write technology allows dielectrics and conductive inks to be printed on the surface in very thin, light weight traces. This technology allows the antenna to add negligible weight and volume to the helmet. Table I shows expected weight values of the components, and the total weight. The cable used in the comparison is a Thermax Item # 710-0201, MaxFlex high-performance flexible cable with 4 feet of cable on the helmet and 3 feet to the radio.

Table I Estimated weight of antenna system on helmet including connectors and cables.

Component	Weight (oz)
4 Printed Antennas	<0.4
On helmet cables	0.83
Connector	<0.15
Cable to Radio	0.62
Total	< 2.0

HUMAN FACTORS

Table II 10 g SAR values at specified frequencies at 500 mW TX power.

Frequency in GHz	Maximum 10 g SAR W/kg
0.75	0.98
0.86	1.32
1.35	0.777
1.5	0.69
1.75	1.05
2	1.32
2.25	0.75
2.5	0.60
2.7	0.70

A major limiting factor of a body worn antenna is RF absorption by the body which can be assessed by the specific absorption rate (SAR) analysis. This limits the amount of power that can be transmitted. The IEEE standard [3] limits SAR to 2 W/kg averaged over a 10 g cube. Modeling using a solid phantom block with the dielectric properties of the brain, it was calculated that transmitting a continuous 500 mW RF signal using the antenna in Figure 3, with 1.5 cm stand-off, led to the SAR values found in Table II. This shows that maximum continuous transmit power is limited to less than a watt. However, normal RF operations have low duty cycles which will allow increasing this level significantly. Additional reduction in SAR values can be achieved through shielding or other methods to increase transmit power to 5 W.

INTERCONNECT

The helmet interconnect is a critical feature of this system. It must be lightweight, have a low profile, and be easy to connect/disconnect with gloved hands. In addition, the tether must be lightweight, rugged and not inhibit the motion of the head in any manner. The connector we have chosen is a MOLLE-type (Modular Lightweight Load-carrying Equipment) RF connector. In particular we are investigating using the Snapnet™ connector that has been developed for the Army's Natick Soldier Center by Physical Optics Corp. (POC, Torrance, CA).

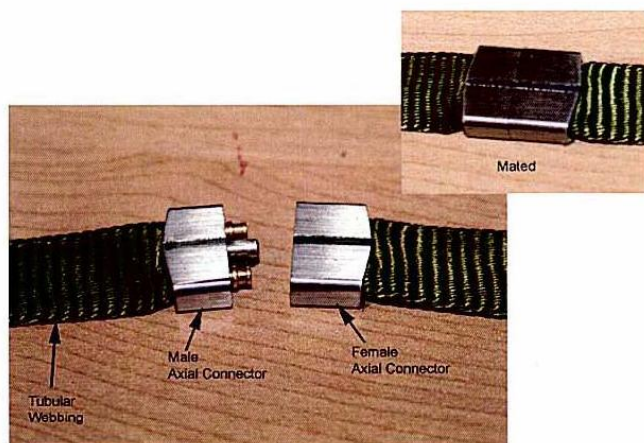


Figure 2. Physical Optics Corp.'s Snapnet™ Connector

The Snapnet™ connector (Figure 2) is based on the GPPG Corning Gilbert "blindmateable adaptor" and can be configured with multiple conductors for signals from DC to 65GHz with a low Voltage Standing Wave Ratio

(VSWR) in the regions of interest. The connectors mount directly onto a rigid structure with circuit traces eliminating the need for any additional webbing on the helmet. These connectors are modeled after standard snaps on MOLLE frames and are extremely user friendly. They have a 0.59 inch diameter and measure 0.16 inches high, and each weighs approximately 0.15 oz. The connectors meet MIL-PRF-39012 environmental standards. The male connector mounts to the lower edge of the helmet in the back center with printed transmission lines on the helmet connecting them to the antenna(s).

The female connector provides interconnect to the radio(s) with the cabling encased in tough, flexible, abrasion-proof webbing (used in parachutes). Depending on the immediate application (tactical radio, GPS, etc.) the harness can be terminated with the appropriate 50-ohm connector.

MANUFACTURING

There are several manufacturing challenges associated with the integration of conformal, direct written antennas into ground combat helmets. Current generation ballistic helmets are comprised of aramid fabrics impregnated with polyvinylbutyral (PVB) modified phenolic resin and consolidated under high temperature and pressure. Integration of the antenna could occur following the molding operation, but prior to finishing of the shell (beading, painting, and final assembly). Such a process would be directly applicable to retrofitting existing helmets in the field (since they would be pre-molded), and would prevent the antenna from being subject to the high temperature and high pressure molding cycle used in current helmet manufacturing. This is a relatively low risk integration option, imparting challenges associated with antenna-paint compatibility, EMI shielding integration, and antenna durability. Durability is considered the most significant challenge, because applying the antenna to the external surface of the helmet potentially exposes the antenna to environmental effects and low velocity impact during normal use and wear of the helmet system.

The durability of the antenna could be greatly improved if the antenna could be written on the preimpregnated aramid fabric prior to helmet consolidation and processed through the temperature and pressure cycle associated with helmet molding. In this case, the antenna could be located within the thickness of the helmet shell, completely protected from low velocity impact, environmental effects, and other potential

exposures during use. The current PVB modified phenolic resin cures via a condensation reaction, generating water and phenol formaldehyde byproducts during the cure process. The effect of such byproducts on the antenna materials is yet unknown.

Next generation ballistic helmet technologies include the use of thermoplastic resin systems that may alleviate the exposure to water and phenol formaldehyde byproducts, thermoforming or other rapid molding processes that may decrease or eliminate the elevated temperature and high pressure exposure during helmet consolidation, and hybrid fabrics that may facilitate the incorporation of EMI shielding, ground planes, or other electrical features directly into the design of the helmet, simplifying the integration of direct written antennae into helmet system. These technologies offer the ability to reduce or eliminate many of the potential challenges associated with current processes, but would not allow for the retrofit of antennas into currently fielded helmets.

The key technology that allows us to integrate the antenna within the helmet itself is the conductor/antenna fabrication process itself. Ink jet and micropen dispensing are perhaps the oldest, most established Direct Write techniques. For both techniques, a conductive material (e.g., silver or nickel based ink) or dielectric material is dispensed onto the substrate in liquid form. A piezoelectric transducer is used for ink jet dispensing low-viscosity inks, and a motorized micropen is used for higher-viscosity inks. In both cases, the material is dispensed onto the ply through a narrow orifice in liquid form at the desired resolution. Most conductive PTF inks are composed of silver particles dispersed in a non-conductive polymer binder with a suitable solvent. Heat is required to drive the solvent out of the printed ink (causing the ink to solidify) or to bring about chemical crosslinking of the binder (e.g., epoxy). The non-conductive binder imparts desirable mechanical properties to the ink, ensuring that the ink retains its cohesive integrity and its adhesion to the substrate. The binder generally does not support electrical current flow, however. Removal of solvent from the ink and binder curing brings the conductive particles into physical contact so as to support electrical current. UV-curable conductive inks are also available, with minimal heat input into the composite part. The "circuitized" composite layers are then assembled into a fully cured ballistic composite for testing.

Some advantages of this approach are:

1. Space and weight savings due to elimination of external antennas and their mechanical support structures.

2. Increased covertness, since there are no external antennas indicating the wearer is a communication specialist.
3. Increased reliability and survivability due to encasing the system within the ballistic helmet itself.
4. Single, blind-mate, low profile connector riveted to the back of the helmet is hardly noticeable at any distances.

PROTOTYPE AND TEST RESULTS

To validate the computational antenna model, a prototype was built on a Rogers 5880 substrate as shown in Figure 3. An impedance measurement was taken, and a very good correlation between simulated and measured return loss was found as shown in Figure 4.

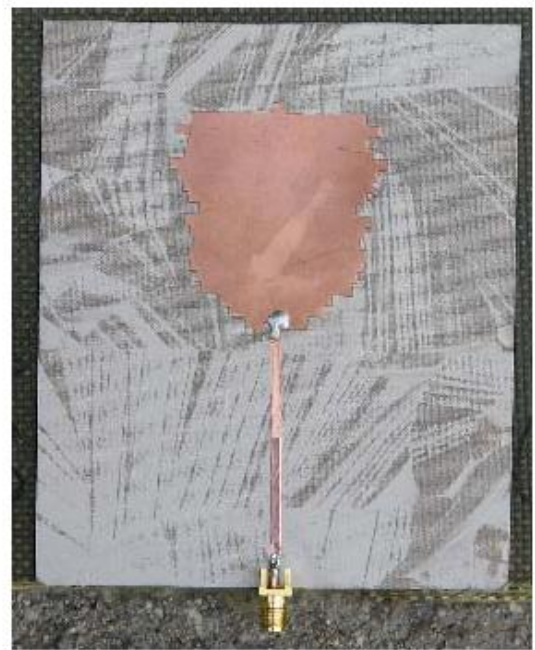


Figure 3 Prototype antenna built on Rogers 5880.

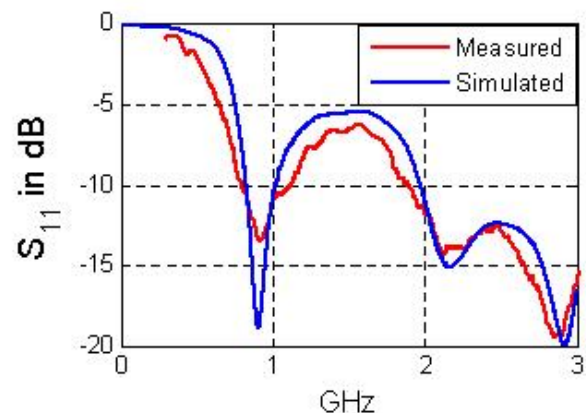


Figure 4 Measured and simulated return loss of prototype antenna built on a PCB.

With confidence in the computer model, additional optimization was done on the antenna reducing the return loss over the bands of interest. These designs were then printed onto a Kevlar substrate similar to those used for ballistic helmets.



Figure 5 Optimized dual band design covering 750-900 MHz and 1350-2700 MHz.

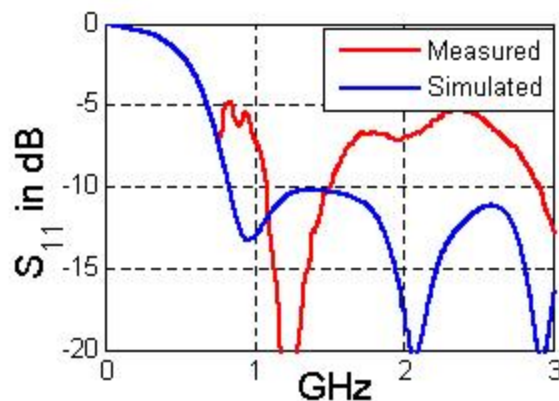


Figure 6 Measured and simulated return loss of dual band direct write antenna.

Figure 5 shows the optimized design, and Figure 6 shows its measured and computed return loss. The design consists of several fixed height strips that make up the length of the antenna. The width of each strip is optimized to give a good broadband performance. By optimizing the width of each strip and the height, very good broadband results can be achieved. Measured performance isn't as good as simulated performance. This is likely a result of the impedance of the micro-strip

transmission line. Additional characterization of the dielectric needs to take place to ensure the impedance of the micro-strip line is 50 Ohms. The impact of the adhesive also needs to be assessed.



Figure 7 Optimized high band design covering 1350-2700 MHz.

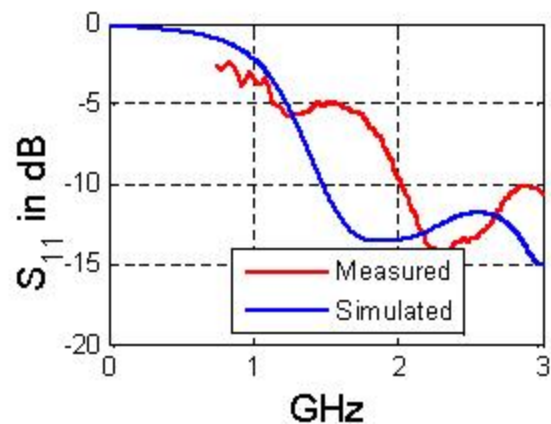


Figure 8 Measured and simulated return loss of high band direct write antenna

An additional antenna was designed and tested that was designed to operate over the 1350-2700 MHz band as shown in Figure 7. Its simulated and measured performance is shown in Figure 8. Again, the return loss is worse than expected. Again, this is likely a result of the impedance of the micro-strip transmission line and possible effects of the adhesives used.

Simulated antenna patterns are fairly omni directional in the horizontal plane. It is expected that the pattern will change significantly when placed on a helmet. The human head reflects/absorbs significant energy and will

tend to direct the pattern away from the head. This will be validated when measurements can be taken with the antennas printed on the curved helmet.

CONCLUSIONS

An Advanced Combat Helmet has been successfully upgraded with a broadband antenna that gives coverage over the 750MHz through 2.7GHz frequency band. The helmet is currently at CERDEC for further test and evaluation.

The preliminary test and manufacturing results of the prototype effort are very encouraging. Some adjustments in the dimensions of the antenna are required to account for the compound curvature effects of the helmet. Additional efforts in the deposition of the "micro-strip" transmission line and the connectorization of the antennas are also being looked into.

A path forward has been defined for the program and currently the helmet is planned to be demonstrated over the summer of 2007 with some field trials. Helmet mounted antennas promise to provide both light-weight and covert solutions for the foot-soldier of the near future.

FUTURE WORK

US Army RDECOM CERDEC will conduct additional evaluation of the antenna design. SAR and antenna gain measurements of the helmet antenna prototype on manikin, fill with CTIA brain-simulating liquid [4], will be conducted at CERDEC laboratory facilities.

Future efforts on this program will be focused on further reducing the weight carried by the soldier on his head. Utilizing the unique direct writing of conductive patterns onto the helmet as demonstrated in the antenna implementation, circuit traces can also be drawn. Another technology developed at SI2 is that of Laser Transfer (Patent Pending), whereby passive and active circuit components are transferred to doubly-curved surfaces. This technology eliminates the need for traditional mechanical structures such as housings and circuit boards and allows the electronics to be embedded directly within the plies of the helmet itself. Applications being investigated include the integration of communication headset electronics.

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